

Gunter & Scott 1990 — Inventory of Definitions and Theorems

Source: **C. A. Gunter and D. S. Scott**, “**Semantic Domains**,” *Handbook of Theoretical Computer Science* Vol. B, North-Holland, 1990, pp. 633–674 ([../papers/Gunter Scott 1990.pdf](#)).

The work list for the Lean formalization: every definition and every one of the paper’s **30 numbered results** (Theorems / Lemmas / Proposition 1–30), in paper order, matched to its Lean equivalent.

Progress (as of r0031, 2026-0806)

#	Quantity	Done	Remaining	Of
1	Definitions to define	all ≈ 13 — the computable function landed in r0031; see the note below on D^∞	0	≈ 13
2	Numbered results complete	17 (Thm 1, Thm 3, Lem 4, Lem 5, Thm 6, Thm 7, Lem 8, Lem 10, Thm 11, Lem 13 , Prop 15, Lem 17, Lem 19, Lem 20, Thm 21, Thm 22, Lem 23)	12	29
2a	— of which partially proved	Thm 16 (the algebraic-lattice conjunct proved; the $Fp(D) \hookrightarrow (D \rightarrow D)$ embedding conjunct not stated)	—	—
3	Unnumbered prose claims proved	12	—	—
4	Mathlib foundations reused	12	—	12
5	Theorems in the development	496 live (+6 commented out as unused)	—	—
6	sorry in the development	—	1 : thm18 in Skeleton/Section6.lean	—

Round r0029 ran four agents in parallel and closed the definition list.

#	Stream	Landed
1	Powerdomain/Hoare.lean	the Hoare (lower) powerdomain: Pf the finite non-empty subsets of $K(D)$, the lower pre-order, and Domain from Theorem 11
2	Powerdomain/Smyth.lean	the Smyth (upper) powerdomain, same shape, dual orientation
3	Powerdomain/Plotkin.lean	the Plotkin (convex) powerdomain under the Egli–Milner pre-order
4	RecursiveDomain.lean	the recursive domain equation, two formalizations of <i>universal domain</i> , and Theorem 21 — plus recursiveDomain_funSpace, the reflexive domain $D \equiv (D \rightarrow D)$

There is no D^∞ to build. Earlier drafts of this inventory listed D^∞ (inverse limit) as an outstanding definition. Reading §7 directly refutes that: the section raises the chain $T_0 \rightarrow e_0 \ T_1 \rightarrow e_1 \ T_2 \rightarrow \dots$, says “This is all very informal, however; how are we to make this idea mathematically precise...?”, and answers with §7.1, *Solving domain equations with closures*. $D \equiv D \rightarrow D$ is reached from Theorem 21 and Lemma 23, and the limit is taken inside $Fc(U)$. What stands in D^∞ ’s place is `Recursive.Solves / IsSolvable`, and the two formalizations of universal domain — `IsUniversal` (every domain of the class is a closure of U) and `IsUniversalRetract` (every one is a retract), which the paper states as two different sentences.

Pf is the finite *non-empty* subsets. The paper defines $Pf(S)$ that way and reserves $\bar{P}f(S)$ for the version including $\emptyset$; all three powerdomains are built over Pf . The distinction is load-bearing and in opposite directions for the three orderings: under the Hoare order $\emptyset \sqsubseteq v$ holds vacuously, so admitting $\emptyset$ would add a point strictly below $\{1\}$; under the Smyth order $\emptyset$ is a **top**, so it would add a spurious maximum; under Egli–Milner $\emptyset$ is comparable to nothing but itself, destroying `OrderBot`. The orchestrator’s brief said “finite subsets”; each agent read the PDF and corrected it.

Namespace per agent worked. Every r0029 declaration lives in `ScottDomains.{Hoare, Smyth, Plotkin, Recursive}`. Four agents, **zero** name collisions — against two in r0028, when five agents shared one namespace.

Round r0031 ran four agents and closed the definition list.

#	Stream	Landed
1	<code>ComputableFunction.lean</code>	the paper’s computable function , on Mathlib’s REPred — the last definition
2	<code>Powerdomain/BoundedComplete.lean</code>	Lemma 13 in the paper’s own wording, and a refutation: see the <code>idealSup</code> defect below
3	<code>Powerdomain/Universal.lean</code>	<code>isRepresentable_prod</code> — the product operator is representable over $P \ N$, the hypothesis
4	Theorem 18	Lemma 24 lacked ; plus $D \equiv D \times D$ still running at the time of writing

The `idealSup` defect — the same class, a third time. `IdealCompletion.idealSup` branches its dte on `Order.IsIdeal (|j| ...)`. That is the membership condition for the *union*, but the union is the least upper bound only when the family is **directed**; for a merely **bounded** family the least upper bound is the ideal *generated by* the union. So `sSup` returns $\perp$ on a bounded non-directed family — kernel-checked in $P \ N$ with the incomparable compacts $\{0\}$ and $\{1\}$, bounded above by $\{0, 1\}$. This follows `ScottHom` (r0006–r0011) and `Smarsh` (r0027), and the r0031 plan asserted the guard was correct, which was wrong. Consequently `not_boundedComplete_hoare`, `_smyth` and `_plotkin` are proved: **no `BoundedComplete` instance can be claimed for any ideal completion until `idealSup` is repaired**, by branching on the *generated* ideal rather than on the union being one.

Lemma 13 is nonetheless true and proved, in the paper’s own wording (`BddAbove S → ∃ I, IsLUB S I`, with the least element from `OrderBot`): D_b (Hoare) needs only the union, so it is bounded complete for **every** domain and the hypothesis is never consumed; $D_\#$ (Smyth) spends both hypotheses. There is no `lem13_plotkin` to prove — Lemma 13 names only $D]$ and $D[$, and Egli–Milner has no join for a bounded pair.

Two inventory rows below are corrected by r0031’s reading of §7. Lemmas 28 and 30 are **not** about $P \ N$ and not about $Fc(U)$: both are stated for *p-representability* over $Fp(U)$, Lemma 28 over §7.3’s ideal completion of dyadic half-open intervals and Lemma 30 over §7.4’s bifinite universal domain. Lemma 23 is therefore **not** Lemma 28’s function-space conjunct. Also, the paper proves $F(X) = X + X$ has **no** representation over $P \ N$, so a $+$ conjunct there is false, not missing.

Theorem 12 is recoverable after all. The row below says its “axioms T ” are never legible; `pdftotext -layout` extracts them cleanly — a continuous algebra $\oplus : E \times E \rightarrow E$ with associativity, commutativity and idempotence, where $T_\#$ adds $s \oplus t \sqsubseteq s$ and T_b adds $s \sqsubseteq s \oplus t$. It belongs to §5.3 with Lemma 13, not to the §7 group.

Round r0028 ran five agents in parallel and roughly doubled the development: $27 \rightarrow 33$ modules, $4440 \rightarrow 8212$ lines, $199 \rightarrow 384$ theorems, $9 \rightarrow 15$ numbered results. Every new result was kernel-audited with `#print axioms`: all depend only on `propext`, `Classical.choice`, `Quot.sound`, and none on `sorryAx`.

#	r0028 stream	Landed
1	CoalescedSum.lean, Skeleton/Sum.lean	$D + E$ as a cpo (sumSup, sumCpo), then lem10_sum, lem17_sum, lem17_smash — Lemma 10 at 6 of 6 conjuncts, Lemma 17 at 5 of 5
2	IdealCompletion.lean	Theorem 11 , both halves, on Mathlib's Order.Ideal; §5 unblocked
3	FinitaryProjectionPoset.lean, Skeleton/Section6b.lean	$F_p(D)$ and $F_c(D)$ as posets, Theorem 16 (algebraic-lattice conjunct), Lemma 20 , and IsClosure.domain_range — Lemma 19 at the paper's strength
4	MinimalUpperBounds.lean	minimal upper bounds, U, U^∞ , and isPlotkinOrder_iff_mubClosure — a characterization the paper does not state. Theorem 18 still open
5	UniversalDomain.lean	Theorem 22 and Lemma 23 , with <i>representable</i> defined from §7; opens the route to D^∞

A name clash lake build could not catch. Streams 3 and 5 each defined IsClosure.apply_sSup_of_directed and isClosure_sSup. The build passed at 971 jobs because no module imported both; the clash appeared the moment anything did — here, an axiom audit importing the pair. isClosure_sSup was the *same* statement proved twice; the single copy now lives in Skeleton/Section6.lean beside IsClosure, which both modules already import, so no call site changed. The two apply_sSup_of_directeds were *different* statements sharing a name — one indexed by a set of the subtype $\mathfrak{r}(im\ r)$, one by an ambient $D : Set\ \alpha$ with $D \subseteq im\ r$ — so the subtype form was renamed IsClosure.apply_sSup_val_image_of_directed. **A green build is not evidence that parallel work composes;** importing every new module together is.

Row 5 counts lines matching $^(@[...])?(theorem|lemma)$ across the 37 modules.

The sorry burn-down (from r0026). The sorrys are deliberate scaffolding: fixed *statements* of outstanding results, confined to ScottDomains/Skeleton/, one file per agent worktree, so that three agents can prove them in parallel without any agent editing a declaration another depends on. Every other module remains sorry-free, and the count above is the burn-down metric — it goes $10 \rightarrow 0$. Round r0027, run as three agents in parallel, took it **10 $\rightarrow$ 1**.

#	Open statement	Result	State after r0027
1	prop15	Prop 15 — every bounded complete domain is bifinite	proved
2	thm18	Thm 18 — $D, D \rightarrow D$ domains $\implies D$ bifinite	sorry — the paper gives no proof, citing Smyth [Smy83a]; the obstacle is recorded in the docstring
3	lem19	Lem 19 — the image of a closure is a domain	proved , via IsClosure.rangeCompletePartialOrder

#	Open statement	Result	State after r0027
4–7	lem10_prod, lem10_smash, lem10_lift, lem10_strict	Lem 10 — bounded completeness closed under $\times, \otimes$, ($\perp$), $\rightarrow \perp$. The $\rightarrow$ conjunct is already proved (Thm 7’s bounded- complete half, r0007)	all four proved ; Lem 10 is then 5 of 6 conjuncts — $D + E$ is not stated
8–10	lem17_prod, lem17_lift, lem17_fun	Lem 17 — bifiniteness closed under $\times$, ($\perp$), $\rightarrow$	all three proved ; Lem 17 is then 3 of 5 conjuncts — $D \otimes E$ and $D + E$ are not stated

The + conjuncts, closed in r0028. CoalescedSum.lean was 181 lines of ingredients with no sSup and no cpo instance, so there was no $D + E$ to state a conjunct over. It now has both, and the guard is the membership condition IsNonBotSum (sumCandidate (sumBase s)) — the defining predicate of the subtype — not directedness, so the defect fixed in ScottHom and then in Smash did not recur a third time. One wrinkle the smash did not have: $\alpha \otimes \beta$ carries no SupSet, so a summand must be selected first; that selection is not a second guard, because a set with an upper bound at all lies in one summand.

The smashSup defect (r0027). lem10_smash was not merely open: as smashSup stood, it was **false**, and the kernel confirmed a refutation. smashSup branched its dite on the base being nonempty *and directed*, so a merely **bounded** non-directed base fell to the adjoined $\perp$, which is not even an upper bound. Witness: $D = \text{Prop} \times \text{Prop}$, $E = \text{Prop}$, $s = \{\uparrow((\text{True}, \text{False}), \text{True}), \uparrow((\text{False}, \text{True}), \text{True})\}$, bounded by $\uparrow((\text{True}, \text{True}), \text{True})$. This is the same defect ScottHom.lean records having already hit for the function space. The repair branches on the coordinatewise supremum landing in NonBotPair — the condition under which it is an element of $D \otimes E$ at all, rather than a merely sufficient condition for it. smashSup_of_directed and smashSup_of_empty kept their statements and were reproved, which is the agreement claim stated in Lean rather than in prose, and smashCpo needed no change.

Kernel check on the r0027 merges (#print axioms): all ten proved statements plus smashCpo, smashSup_of_directed and smashSup_of_empty depend only on propext, Classical.choice, Quot.sound — lem10_prod and lem19 do not even need Classical.choice. None depends on sorryAx. thm18 does, as its sorry requires.

Row 2 counts only the paper’s 30 **numbered** results (Theorems / Lemmas / Proposition 1–30). Row 3 counts the claims the paper makes **in prose** rather than as numbered results; these are paper content too, and all twelve are formally verified:

#	Paper claim	Where	Lean
1	“the compact elements [of $P \rightarrow N$] are just the finite subsets of N ”	p. 9	isCompactElement_iff_finite
2	“ $P \rightarrow N$... is a domain”	p. 9	instance : Domain (Set X)
3	“ $D \rightarrow E$ is a ... cpo”	Thm 7 proof	instance : CompletePartialOrder (ScottHom $\alpha \beta$)
4	“... bounded complete ... whenever E is”	Thm 7 proof	instance : BoundedComplete (ScottHom $\alpha \beta$)
5	“step(s) ... is continuous”	Thm 7 proof	scottContinuous_stepFun
6	“... and compact in the ordering on $D \rightarrow E$ ”	Thm 7 proof	isCompactElement_step
7	“they form a basis for $D \rightarrow E$ ”	Thm 7 proof	instance : IsAlgebraic (ScottHom $\alpha \beta$)
8	every compact function is a <i>finite</i> join of step functions	Thm 7 proof, implicit	exists_finite_isLUB_of_isCo

#	Paper claim	Where	Lean
9	“an embedding is an injection”	§3.1	IsEmbeddingProjectionPair.i
10	“a projection is a surjection”	§3.1	IsEmbeddingProjectionPair.s
11	“it is easy to check that $p_N \dots$ is a finitary projection”	§3.1	isFinitaryProjection_normal
12	“the set of strict continuous functions $D \rightarrow E$ is also a cpo”	§2.1	ScottDomains.strictHomCpo

Six of those eleven are the body of **Theorem 7**, which is now **complete** — all four conjuncts of its conclusion (cpo, bounded complete, algebraic, countably based) are formally verified, as `ScottHom.isBoundedCompleteDomain_scottHom`.

It is proved under **weaker hypotheses than the paper states**: bounded completeness of D is never used. D need only be a domain. The function space is a cpo for any preordered D , algebraic when D and E are, bounded complete because E is, and countably based because D and E are.

The remaining theorems are supporting API: the $\ll$ calculus, the `compactsBelow` machinery, the pointwise order and suprema on $D \rightarrow E$, and the step-function adjunction. The paper assumes or elides all of it.

Reference audit (r0020). Of the theorems in the development, 16 are never cited elsewhere. Nine of those are *terminal by design* — they are the paper’s own claims, so nothing should cite them (`injective_embedding`, `surjective_projection`, the `isNormalIn_sUnion*` family and `mono_right` for Lemma 4, `singleton_bot_isNormalIn` for $\{\perp\} \in P(C)$, `isLeast_kleeneFix_le`, `eq_kleeneOperator_op`). The other six were speculative API written for callers that never appeared; they are **commented out in place**, each with a note on why it exists and what is instructive about it, and the build is unchanged — which also confirms that the three `@[simp]` ones among them were never firing implicitly.

The development is **40 modules, 10779 lines, 1 sorry, 0 other warnings**. Counts of definitions, results and theorems are in the Progress table above — they are not repeated here, so that this section cannot drift out of step with it. What each round delivered:

#	Round	Module	Contents
1	r0003	ScottDomains/WayBelow.lean	way-below $\ll$ at $[Preorder\ \alpha]$, 7 theorems; $x \ll x \leftrightarrow IsCompactElement\ x$ holds by <code>Iff.rfl</code>
2	r0004	ScottDomains/Domain.lean	<code>IsAlgebraic</code> , <code>Domain</code> (with the paper’s countable-basis condition), <code>BoundedComplete</code> , 8 theorems, <code>Domain Prop</code>
3	r0005	ScottDomains/Powerset.lean	the paper’s $P\ \mathbb{N}$ (p. 9): compacts of <code>Set X</code> are exactly the finite subsets, hence <code>Domain (Set ℕ)</code> — the nondegenerate witness
4	r0006–r0007	ScottDomains/ScottHom.lean	the continuous function space $D \rightarrow E$: <code>ScottHom</code> , the pointwise order, <code>CompletePartialOrder</code> , and <code>BoundedComplete</code> when E is — Theorem 7’s first sentence in full
5	r0008	ScottDomains/StepFunction.lean	the single step function <code>step k e</code> : continuity (from k compact), the adjunction <code>step k e ≤ f ↔ e ≤ f k</code> , and compactness in $D \rightarrow E$ (from e compact)
6	r0009	ScottDomains/FunctionSpaceDomain.lean	IsAlgebraic — the paper’s “they form a basis for $D \rightarrow E$ ”; the two halves of <code>IsAlgebraic</code> use disjoint hypotheses (directedness needs only E bounded complete; the lub needs only D, E algebraic)
7	r0010	ScottDomains/CompactFunctionSpace.lean	every compact function is a finite join of step functions — the finiteness half of the basis claim

#	Round	Module	Contents
8	r0011	ScottDomains/FunctionSpaceComp.lean	Table 1 — $K(D)$ is countable, hence Domain (ScottHom α β) — Theorem 7 complete
9	r0012	ScottDomains/NormalSubposet.lean	lean normal-subposet relation $\triangleleft$ and Lemma 4 , all four parts
10	r0012–r0013	ScottDomains/Projection.lean	embedding–projection pairs, projections, the paper’s “an embedding is an injection, a projection is a surjection”; im(p) is a cpo and finitary projections
11	r0014	ScottDomains/FinitaryProject.lean	Lemma 5 — the compacts of $\text{im}(p)$ are $\text{im}(p) \cap K(D)$ (needs only that p is a projection), and $\text{im}(p) \cap K(D) \triangleleft K(D)$
12	r0015	ScottDomains/NormalProjection.lean	$p_N = \bigsqcup \{y \in N \mid y \sqsubseteq x\}$: continuous, a projection, and $\text{im}(p_N) \cap K(D) = N$ — half of Theorem 6 ’s correspondence, plus order preservation both ways
13	r0016	ScottDomains/Theorem6.lean	Theorem 6 — p_N is finitary, $p_{\{\text{im}(p) \cap K(D)\}} = p$, and the correspondence assembled
14	r0017	ScottDomains/FixedPoint.lean	Theorem 1 — $\bigsqcup_n f^n(\perp)$ is the least fixed point of a continuous f on a cpo. Not Mathlib reuse; see the §2 table
15	r0018	ScottDomains/UniformFixedPoint.lean	Theorem 3 — fix is the unique uniform fixed-point operator; $\downarrow a$ as a cpo
16	r0019	ScottDomains/Product.lean	$D \times E$ as a cpo (the one construction needing no case split) and Lemma 8 parts 1–3
17	r0021	ScottDomains/Currying.lean	Lemma 8.4 — currying, $D \rightarrow (E \rightarrow F) \cong (D \times E) \rightarrow F$, and with it Lemma 8 complete
18	r0022	ScottDomains/EffectivePresentation.lean	§2.2 effective presentation — the enumeration of the basis with its two decidability conditions
19	r0023	ScottDomains/Lift.lean	the lift D_1 as a cpo, on Mathlib’s WithBot
20	r0024	ScottDomains/StrictHom.lean	the strict function space $D \rightarrow_1 E$ as a cpo — needs no case split, since both branches of ScottHom’s $s\text{Sup}$ are strict
21	r0025	ScottDomains/Smash.lean	§4.3’s smash product $D \otimes E$ — the non-bottom pairs with a new bottom adjoined, as a cpo
22	r0025	ScottDomains/Bifinite.lean	§6.1’s Plotkin order and bifinite domain

§2 and §3 are now complete — the only §3 omission is the paper’s *computable function*, which needs an r.e.-predicate notion Mathlib does not supply.

Dependency structure of the remaining definitions

#	Definition	Prerequisites	Status
1	smash product $D \otimes E$	Domain, OrderBot	✓ done, r0025
2	bifinite / Plotkin order	IsNormalIn ($\triangleleft$, r0012)	✓ done, r0025
3	sum $D + E$	Domain	independent, ready

#	Definition	Prerequisites	Status
4	D_∞	embedding–projection pairs (r0012)	independent, but large (inverse limits)
5	the three powerdomains	the ideal completion (Theorem 11) — <i>not</i> built	blocked ; the three are independent of each other once it exists

Rows 1 and 2 were written as two modules and checked in a single build, which is the parallelism the structure actually admits — the bottleneck is the build-and-fix loop, not the writing.

Next: the sum $D + E$, then Theorem 11 to unblock the powerdomains, then Lemma 9 (the strict analogues of Lemma 8) and Lemma 10 (closure of bounded completeness).

Work counts

- **Reuse from Mathlib — no work (11):** poset, directed, cpo, $\perp$, monotone, continuous, fixed-point operator, Schröder–Bernstein (2), algebraic lattice, product, λ -notation. (Theorem 1 was listed here in an earlier draft and has been removed: Mathlib’s `OrderHom.lfp` is Knaster–Tarski over a complete lattice, not Kleene’s $\bigsqcup_n f^n(\perp)$ over a cpo. It is proved in `ScottDomains/FixedPoint.lean`, so **29** numbered results needed proof, not 28.)
- **Generalize / adapt (4):** compact element (`IsCompactElement` — its *definition* is already stated at `[PartialOrder  $\alpha$ ]` and so applies to a dcpo verbatim; what is `CompleteLattice`-only is every lemma about it, so the dcpo API must be re-proved), sum, lift (`WithBot/Sum`), ideal completion (`Order.Ideal`).
- **Definitions to define — new (≈ 13 ; 4 done in r0003–r0004, 9 remaining):** way-below $\ll$ (**done** — `ScottDomains/WayBelow.lean`), algebraic cpo, **domain**, bounded-complete (**all three done** — `ScottDomains/Domain.lean`), embedding–projection pair, (finitary) projection, normal subposet, effective presentation, smash product, the three powerdomains (Hoare / Smyth / Plotkin), bifinite / Plotkin order, and D_∞ .
- **Theorems to prove (28):** 28 of the paper’s 30 numbered results — Theorems 3, 6, 7, 11, 12, 14, 16, 18, 21, 22, 25, 26, 27, 29; Lemmas 4, 5, 8–10, 13, 17, 19, 20, 23, 24, 28, 30; Proposition 15. Only Theorems 1 & 2 come free from Mathlib.

Bottom line: ≈ 13 definitions to define + 28 results to prove, on top of 12 reused Mathlib foundations. After r0004: **9 definitions + 28 results** remain.

Lean column legend: ✓ reuse Mathlib (name given) · ~ partial (Mathlib has a related or lattice-only version — generalize) · ✕ define / prove (absent from Mathlib v4.32.2, confirmed by grep).

Statements are paraphrased/de-garbled from the PDF (1990 Type-3 fonts render $\rightarrow$ as $!$, drop `fi` ligatures, mangle math), so read them as a guide to *what* each result is. Notation: $\sqsubseteq$ order, $\bigsqcup$ directed sup, $\perp$ bottom, $\ll$ way-below, $K(D)$ compacts, $Fp(D)$ finitary projections.

§2 Recursive definitions of functions

§	Kind	Ref	Concept / statement	In Lean / Mathlib?
2.1	Def	—	poset (partially ordered set)	✓ <code>PartialOrder</code>
2.1	Def	—	directed subset M : every finite $u \subseteq M$ has an upper bound in M	✓ <code>DirectedOn</code> / <code>IsDirected</code>
2.1	Def	—	cpo : poset in which every directed M has a lub $\bigsqcup M$	✓ <code>CompletePartialOrder</code> (chains: <code>OmegaCompletePartialOrder</code>)
2.1	Def	—	bottom $\perp$ (least element)	✓ <code>OrderBot</code>
2.1	Def	—	monotone function	✓ <code>Monotone</code> / <code>OrderHom</code>
2.1	Def	—	continuous : monotone, $f(\bigsqcup M) = \bigsqcup f(M)$ for directed M	✓ <code>ScottContinuous</code>

§	Kind	Ref	Concept / statement	In Lean / Mathlib?
2.1	Thm	1	Fixed-Point Theorem: $f : D \rightarrow D$ continuous $\implies$ least fixed point $\bigsqcup_n f^n(\perp)$	✓ proved (r0017) — ScottDomains.theorem1. Not Mathlib reuse: OrderHom.lfp is Knaster–Tarski over a <i>complete lattice</i> with only monotonicity, a different theorem
2.2	Thm	2	Schröder–Bernstein for sets	✓ Function.schroeder_bernstein (SetTheory/Cardinal/SchroederBernstein.lean — the name in an earlier draft of this row, Function.Embedding.schroederBernstein, does not exist
2.3	Def	—	fixed-point operator (uniform)	✓ OrderHom.lfp / LawfulFix
2.3	Thm	3	The standard operator is the unique uniform fixed-point operator	✓ proved (r0018) — ScottDomains.theorem3; a fixed point operator is formalized as a family over every CompletePartialOrder in a universe

§3 Effectively presented domains

§	Kind	Ref	Concept / statement	In Lean / Mathlib?
3.1	Def	—	compact element $x: x \sqsubseteq \bigsqcup M(\text{dir.}) \implies x \sqsubseteq y$ some $y \in M; K(D)$	~ IsCompactElement — def is [PartialOrder], so reusable on a dcpo; its lemmas are all CompleteLattice-only
3.1	Def	—	way-below $\ll$ (approximation relation behind compactness)	✓ ScottDomains.WayBelow (r0003) — $x \ll x \leftrightarrow \text{IsCompactElement } x$ by Iff.rfl
3.1	Def	—	algebraic cpo: $x = \bigsqcup \{x' \in K(D) : x' \sqsubseteq x\}$ (directed)	✓ ScottDomains.IsAlgebraic (r0004)
3.1	Def	—	domain = algebraic cpo whose basis $K(D)$ is countable (the paper’s definition, p. 9 — the countability condition was missing from an earlier draft of this row)	✓ ScottDomains.Domain (r0004)
3.1	Def	—	bounded complete: $\perp +$ every bounded subset has a sup — a <i>separate</i> predicate; the paper composes them as “bounded complete domain” (Thm 7, Lem 10, Lem 13, Thm 14), which is the literature’s <i>Scott domain</i>	✓ ScottDomains.BoundedComplete (r0004); the compound is [Domain α] [BoundedComplete α]
3.1	Def	—	(countably based) algebraic lattice	✓ IsCompactlyGenerated (+CompleteLattice)
3.1	Def	—	embedding–projection pair (g, f)	✓ ScottHom.IsEmbeddingProjectionPair (r0012)
3.1	Def	—	projection; finitary projection p : $p \circ p = p \sqsubseteq \text{id}$, $\text{im}(p)$ a domain	✓ ScottHom.IsProjection (r0012), ScottHom.IsFinitaryProjection (r0013) — $\text{im}(p)$ carries a CompletePartialOrder via IsProjection.rangeCompletePartialOrder
3.1	Def	—	normal subposet / substructure	✓ ScottDomains.IsNormalIn, notation $\triangleleft$ (r0012)

§	Kind	Ref	Concept / statement	In Lean / Mathlib?
3.1	Lem	4	$\langle P(C), \triangleleft \rangle$ of substructures is a cpo with $\{\perp\}$ least	✓ proved (r0012), all four parts
3.1	Lem	5	p finitary projection $\implies$ compacts of $\text{im}(p)$ are $\text{im}(p) \cap K(D)$, and $\text{im}(p) \cap K(D) \triangleleft K(D)$	✓ proved (r0014) — <code>IsProjection.isCompactElement_iff</code> (needs only <i>projection</i>) and <code>IsFinitaryProjection.isNormalIn_compacts</code>
3.1	Thm	6	Isomorphism: normal substructures $\equiv \text{Fp}(D)$ (finitary projections)	✓ proved (r0015–r0016) — <code>ScottDomains.theorem6</code>
3.1	Thm	7	D, E bounded-complete domains $\implies D \rightarrow E$ bounded-complete domain	✓ proved (r0006–r0011) — <code>ScottHom.isBoundedCompleteDomain_scottHom</code> ; D bounded complete is not needed
3.2	Def	—	effective presentation $d : \mathbb{N} \rightarrow K(D)$; effectively presented domain	✓ <code>ScottDomains.EffectivePresentation</code> (r0022). The paper’s computable function is ✓ defined (r0031) in <code>ComputableFunction.lean</code> , on Mathlib’s REPred (Mathlib/Computability/RE.lean:157). An earlier draft of this row said “Mathlib v4.32.2 has no RePred or equivalent (grep finds none)” — that grep used the wrong capitalization, and the claim was false. Two caveats recorded there: <code>decidableLE</code> is too weak to prove anything computable (a <code>Decidable</code> instance may be <code>Classical.dec</code>), so results need an explicit <code>RecursiveLE</code> hypothesis; and composition awaits <code>REPred</code> closure under $\wedge$ and $\exists$, which Mathlib’s five-lemma API does not supply

§4 Operators and functions

§	Kind	Ref	Concept / statement	In Lean / Mathlib?
4.1	Def	—	product $D \times E$	✓ Prod order from Mathlib; the cpo instance is <code>ScottDomains.instCompletePartialOrderProd</code> (r0019) — Mathlib has <code>Prod.supSet</code> and <code>isLUB_prod</code> but no cpo instance
4.2	Def	—	Church’s λ-notation (continuous abstraction)	✓ <code>OrderHom / ωCPO ContinuousHom</code>
4.3	Def	—	smash product $D \otimes E$	✓ <code>ScottDomains.Smash + smashCpo</code> (r0025)
4.4	Def	—	sum $D + E$; lift $D_\perp$	both ✓ — <code>lift ScottDomains.liftCpo</code> on <code>WithBot</code> (r0023); the coalesced sum <code>CoalescedSum</code> with <code>sumSup</code> and <code>sumCpo</code> (r0028), its <code>sSup</code> guarded on landing in <code>NonBotSum</code>
4.x	Lem	8	$D \times E \equiv E \times D$; $(D \times E) \times F \equiv D \times (E \times F)$; $D \rightarrow (E \times F) \equiv (D \rightarrow E) \times (D \rightarrow F)$; $D \rightarrow (E \rightarrow F) \equiv (D \times E) \rightarrow F$	✓ proved — <code>prodComm</code> , <code>prodAssoc</code> , <code>scottHomProd</code> (r0019); <code>scottHomCurry</code> (r0021)

§	Kind	Ref	Concept / statement	In Lean / Mathlib?
4.x	Lem	9	Product/function-space iso laws over D, E, F	✗ not statable — the PDF drops every $\otimes$ and $\perp$, so which operators the laws range over is unreadable. Statement recovery is r0030 agent5’s assignment
4.5	Lem	10	D, E bounded complete $\implies \rightarrow, \times, \otimes, +, () \perp$ bounded complete	✓ proved, 6 of 6 conjuncts — $\rightarrow$ r0007; $\times, \otimes, () \perp, \rightarrow \perp$ r0027 (Skeleton/Lemma10.lean); $+$ r0028 (Skeleton/Sum.lean)
4.5	Thm	11	Ideal completion of a countable pre-order is a domain (all domains so arise)	✓ proved (r0028) — IdealCompletion.thm11 and thm11_converse, on Mathlib’s Order.Ideal
4.5	Thm	12	Initiality of a continuous algebra satisfying axioms T	✗ prove — statable after all (r0031): the T axioms extract cleanly as a continuous $\otimes$ with associativity, commutativity, idempotence, plus $s \otimes t \sqsubseteq s$ for $T \#$ and $s \sqsubseteq s \otimes t$ for $T \flat$. It sits in §5.3 with Lemma 13
4.5	Lem	13	D bounded complete $\implies$ powerdomains $D], D[$ bounded complete	✓ proved (r0031) — PowerdomainBC.lem13_hoare, lem13_smyth, in the paper’s wording. $D \flat$ needs no bounded-completeness hypothesis at all; there is no convex conjunct to prove
4.5	Thm	14	Equivalent characterizations of an (algebraic/BC) domain	✗ not statable — the list of characterizations is garbled. Statement recovery is r0030 agent5’s assignment

§5 Powerdomains

§	Kind	Ref	Concept / statement	In Lean / Mathlib?
5.1	Def	—	powerdomain (non-deterministic outcomes)	✓ each of the three is IdealCompletion (Pf K(D)) under its pre-order (r0029)
5.2	Def	—	Hoare (lower), Smyth (upper), Plotkin (convex) powerdomains	✓ ScottDomains.Hoare.Powerdomain, Smyth.Powerdomain, Plotkin.Powerdomain (r0029), each with its Domain instance from Theorem 11 and its compacts characterized as the principal ideals
5.3	—	—	Universal & closure properties (see Lem 13, 28, 30)	✗ prove — r0030 wave A

§6 Bifinite (SFP) domains

§	Kind	Ref	Concept / statement	In Lean / Mathlib?
6.1	Def	—	Plotkin order / bifinite (SFP) domain	✓ ScottDomains.IsPlotkinOrder, IsBifinite (r0025)
6.1	Prop	15	Every bounded-complete domain is bifinite	✓ proved (r0027) — ScottDomains.prop15, the paper’s own proof over lubClosure u

§	Kind	Ref	Concept / statement	In Lean / Mathlib?
6.1	Thm	16	D bifinite $\implies \text{Fp}(D)$ is an algebraic lattice	~ algebraic-lattice conjunct proved (r0028) — <code>ScottDomains.thm16</code> . The $\text{Fp}(D) \hookrightarrow (D \rightarrow D)$ embedding conjunct is not stated; the paper’s S_f sketch has a documented gap, and it is r0030 agent4’s assignment
6.2	Lem	17	D, E bifinite $\implies \rightarrow, \times, \otimes, +, ()^\perp$ bifinite (incl. function space)	✓ proved, 5 of 5 conjuncts — $\times, ()^\perp$, $\rightarrow$ r0027 (<code>Skeleton/Lemma17.lean</code>); $\otimes$, $+$ r0028 (<code>Skeleton/Sum.lean</code>)
6.2	Thm	18	If D and $D \rightarrow D$ are domains, then D is bifinite	✗ prove — the development’s only sorry . Reduced by <code>isBifinite_iff_mubClosure</code> (r0028) to two obligations; blocked on a constructor for continuous functions on a domain that is not bounded complete
6.2	Lem	19	closure $r : D \rightarrow D$ ($r \circ r = r \sqcap \text{id}$) $\implies \text{im}(r)$ is a domain	✓ proved — <code>lem19</code> (r0027) gives the cpo structure; <code>IsClosure.domain_range</code> (r0028) gives the paper’s full strength, $\text{im}(r)$ a domain with basis $\{r(k) \mid k \in K(D)\}$
6.2	Lem	20	D domain $\implies \text{Fc}(D)$ (finitary closures) is a cpo	✓ proved (r0028) — <code>ScottDomains.lem20</code> , over $\text{Fc } \alpha$ with the pointwise order

§7 Recursive definitions of domains (universal domain, D_∞)

§	Kind	Ref	Concept / statement	In Lean / Mathlib?
7	Def	—	recursive domain equation; universal domain. (Earlier drafts of this row said “ D_∞ (inverse limit)” — §7 builds no inverse limit; see the note under Progress)	✓ <code>Recursive.Solves / IsSolvable</code> , and <code>Recursive.IsUniversal / IsUniversalRetract</code> for the paper’s two phrasings (r0029)
7	Thm	21	F representable over cpo $U \implies$ a domain D with $D \cong F(D)$	✓ proved (r0029) — <code>ScottDomains.Recursive.thm21</code> ; with <code>IsRepresentable₂.diag</code> and Lemma 23 it yields <code>recursiveDomain_funSpace</code> , the reflexive domain $D \cong (D \rightarrow D)$
7	Thm	22	Any countably-based algebraic lattice L : a closure $r : P(\mathbb{N}) \rightarrow L$	✓ proved (r0028) — <code>ScottDomains.thm22</code> , with <code>thm22_of_isCompactlyGenerated</code> the Mathlib-vocabulary form
7	Lem	23	The function-space operator is representable over $P(\mathbb{N})$	✓ proved (r0028) — <code>ScottDomains.lem23</code>
7	Lem	24	U cpo; $\times$ and $\rightarrow$ representable over $U \implies$ (setup for universality)	✗ prove
7	Thm	25	U non-trivial domain representing $\times, \rightarrow \implies$ U universal	✗ prove
7	Thm	26	Any signature $(s_1, \dots, s_n)$: combinators $F_1, \dots, F_n$ solving the equations	✗ prove
7	Thm	27	Any bounded-complete D : a projection of the universal domain onto D	✗ prove
7	Lem	28	Operators $\rightarrow, \times, \otimes, +, ()^\perp, ()^\perp, ()^\perp [$ representable over U	✗ prove

§	Kind	Ref	Concept / statement	In Lean / Mathlib?
7	Thm	29	D bifinite $\implies D^+$ bifinite; solving $D \cong D^+$	✗ prove
7	Lem	30	Operators $\rightarrow, \times, \otimes, +, ()^\perp, ()^\top, ()^\bot$ p-representable over universal bifinite V	✗ prove

Tally (matched): ✓ reuse Mathlib $\approx$ **12** (poset, directed, cpo, $\perp$, monotone, continuous, fixed-point Thm 1 & operator, Schröder–Bernstein, algebraic lattice, product, λ -notation) · $\sim$ partial $\approx$ **4** (compact element, sum/lift, ideal completion Thm 11, Thm 22) · ✗ define / prove $\approx$ **44**, of which 4 are done ($\ll$ r0003; algebraic, domain, bounded-complete r0004) $\rightarrow$ **40 remaining** — the bifinite / powerdomain / D^∞ development and all 28 numbered results.

What is done, and what is next, is listed under **Progress** at the top of this file.